wingfoil

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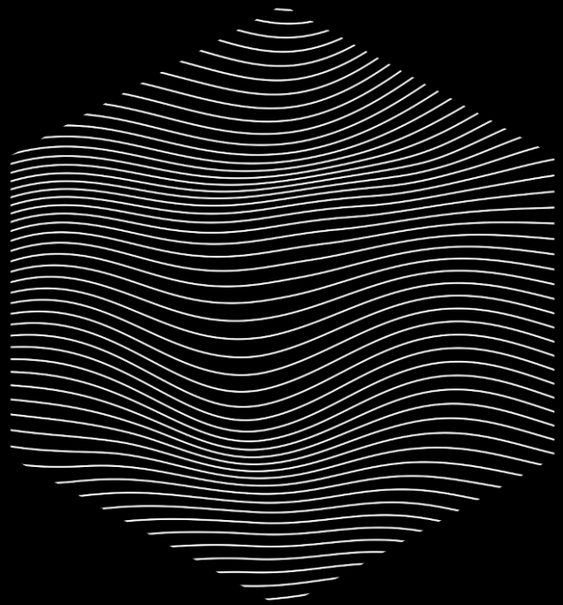
the ultra low latency data streaming framework

August 2026

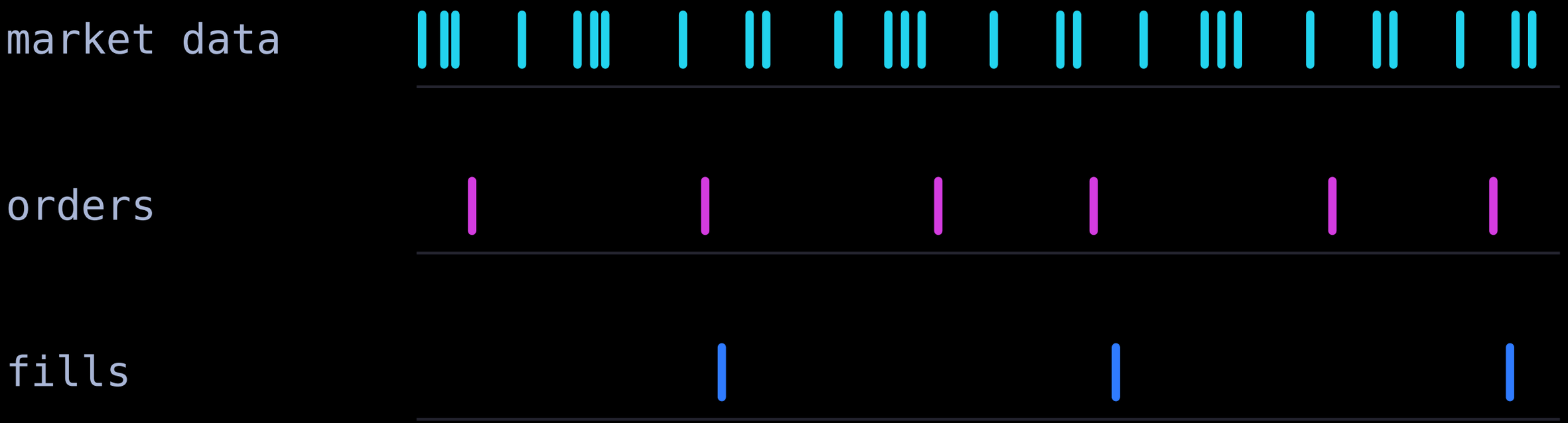


Jake Mitchell

25 years in financial services · JPMorgan · B2C2

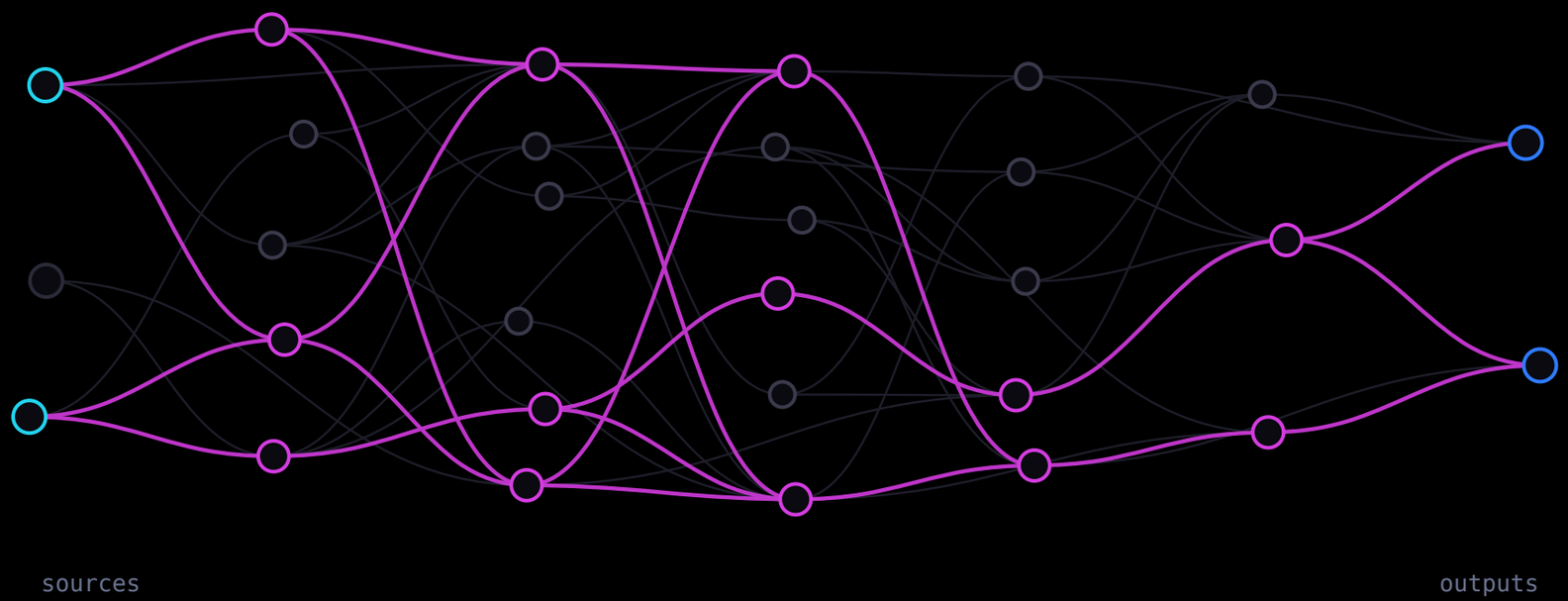
THE PROBLEM

Everything ticks at a different rate



THE SOLUTION

Wire a graph of calculations



- The framework abstracts over **what needs to tick, and when**.
- **Split and recombine** freely — it stays one graph.
- **Modulate the frequency** — window, throttle, sample.
- And it has to be **efficient**.

THE SURFACE

Wire it, build it, run it

```rust
let g = GraphBuilder::new();

let printed = g
    .ticker(Duration::from_millis(100))
    .count()
    .map(|i| format!("tick {i}"))
    .for_each(|msg: &String| { println!("  {msg}"); Ok(()) });

g.build().run(RunMode::HistoricalFrom(NanoTime::ZERO), RunFor::Cycles(5))?;
```

```
historical run (instant):
  tick 1
  tick 2
  tick 3
  tick 4
  tick 5
```

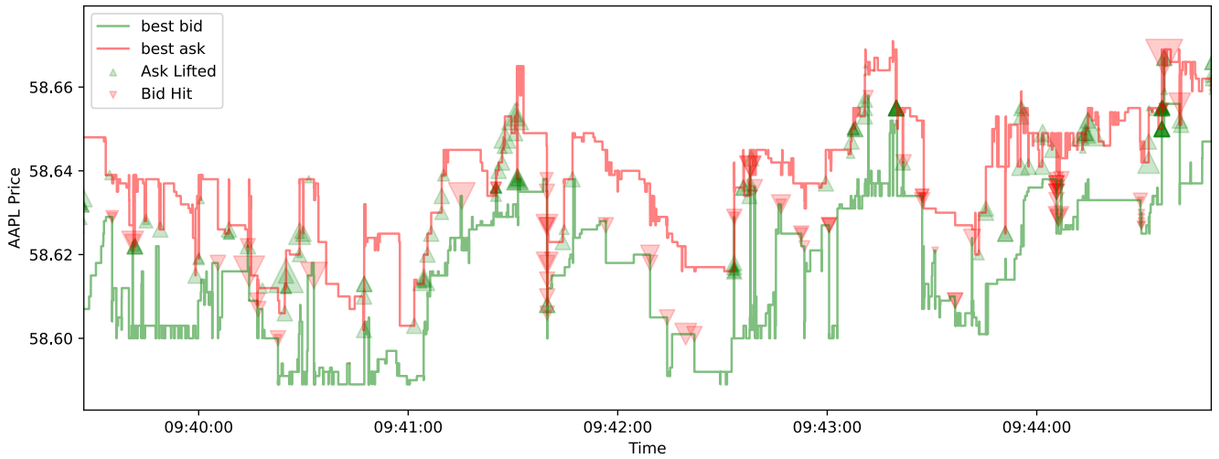
```
cargo run -p wingfoil --example hello_graph
```

MOTIVATION, NOT THESIS

A real graph: NASDAQ limit orders → book → prices + fills

```
let (fills, prices) =
    csv_read(&g, &src, get_time, true, None)?
        .map(move |chunk: &Burst<Message>| {
            process_orders(chunk, &book)
        })
        .split();

prices.filter_none().distinct()
      .csv_write(&prices_path)?;
fills.csv_write(&fills_path)?;
```



RUN MODES

Backtest it, then run it live

`RunMode::HistoricalFrom(NanoTime::ZERO)` — the whole hour

```
$ cargo run --release --example top_of_book
0.021_311  bid  585.33  ask  585.91  spread  0.58  mid  585.62
0.197_502  bid  585.33  ask  585.92  spread  0.59  mid  585.62
0.197_540  bid  585.33  ask  585.93  spread  0.60  mid  585.63
0.201_332  bid  585.36  ask  585.93  spread  0.57  mid  585.64
0.267_498  bid  585.73  ask  585.74  spread  0.01  mid  585.74
```

`RunMode::RealTime` — a three-second window of the same file · animation slowed 12×

```
$ cargo run --release --example top_of_book -- realtime
1,787,777,083.871_535  bid  585.33  ask  585.91  spread  0.58  mid  585.62
1,787,777,084.048_929  bid  585.33  ask  585.92  spread  0.59  mid  585.62
1,787,777,084.049_141  bid  585.33  ask  585.93  spread  0.60  mid  585.63
1,787,777,084.053_476  bid  585.36  ask  585.93  spread  0.57  mid  585.64
1,787,777,084.119_906  bid  585.73  ask  585.74  spread  0.01  mid  585.74
```

I/O

Sixteen adapters, three patterns

MESSAGING
Kafka · ZeroMQ · Fluvio · Redis · etcd

LOW-LATENCY TRANSPORT
Aeron · iceoryx2

TRADING
FIX

STORAGE & FILES
kdb+ · PostgreSQL · CSV · lines

WEB
WebSocket client · WebSocket server

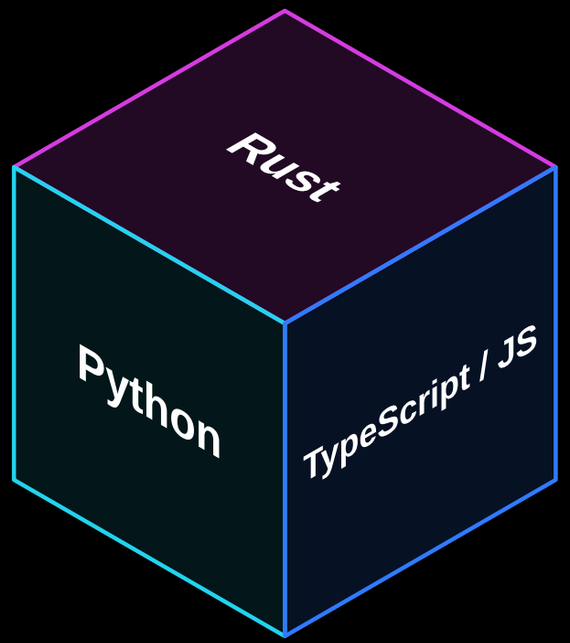
TELEMETRY
Prometheus · OpenTelemetry

Pattern	Latency	Example
Busy loop	lowest	`iceoryx2`
Worker thread	one channel hop	`Kafka`
Async	a hop and a task wake	`PostgreSQL`

Mix and match — the core stays synchronous either way, so **you are not locked into async**.

SURFACES

Three languages, one engine



PART TWO

Nitro

Performance, the wall we hit chasing it,
and the pattern that got us through.

THE HOOK

127 levels deep. 2^{127} distinct paths. One tick.

```
let mut source = g.constant(1_u128);
for _ in 1..128 {
    source = source.join(&source, |a, b| a + b);   // branch and recombine
}
```

```
1 ticks processed in 12.953µs
value 170141183460469231731687303715884105728
```

That value is 2^{127} — the correct answer, which is also exactly the number of source→sink paths a framework that propagates *one path at a time* would have to walk.

WHY IT STAYS FLAT

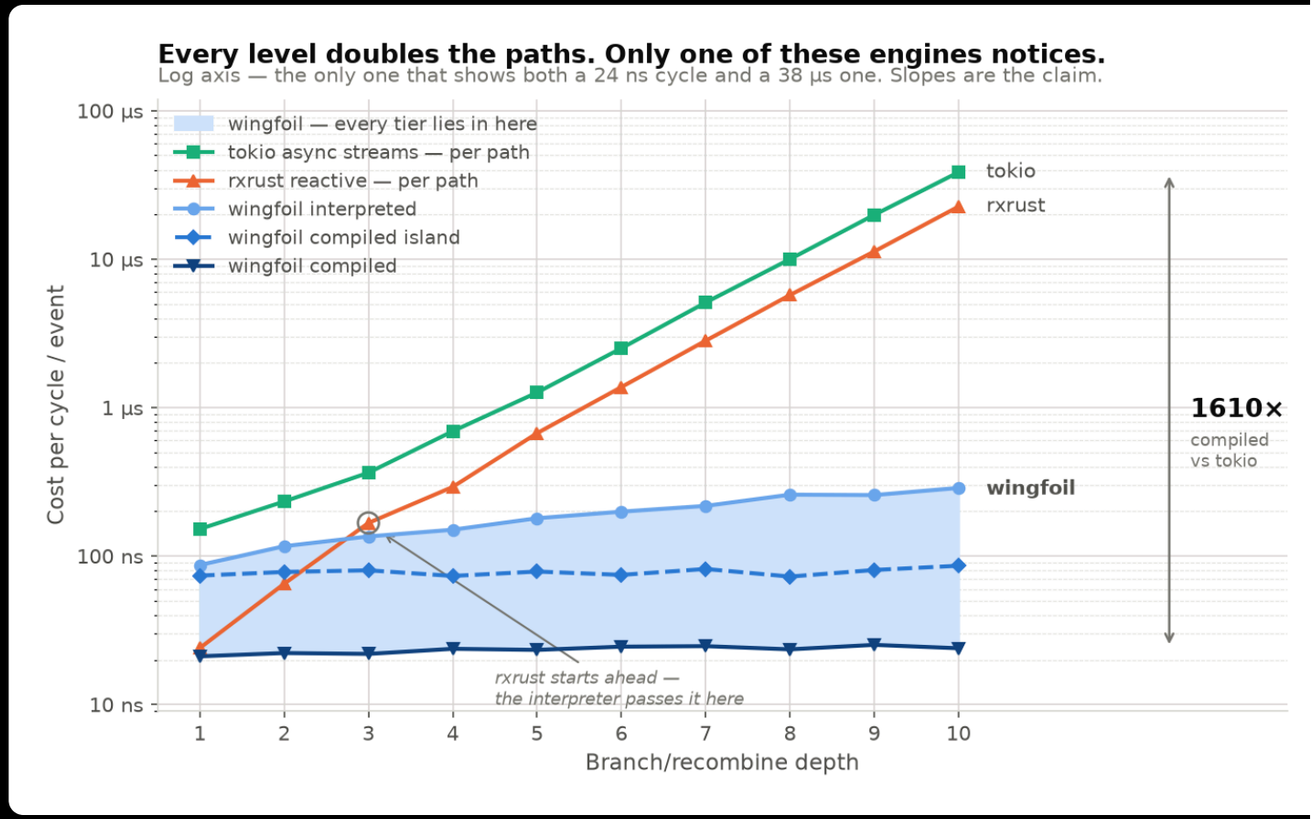
Visit each node once per tick

A node runs **after everything it reads**, exactly once — however many paths lead to it.

Propagate one path at a time and a shared node is re-visited per path. Branch and recombine, and that **doubles every level**.

Rx · async streams

Measured slopes 2.01× and 1.94× per level. Wingfoil: flat.



THE ENGINEERING PROBLEM

What we wanted: compile the graph

Interpreted, every node pays a **dynamic dispatch** and a **trip through a value slot** — per node, per cycle.

On a hot chain that dominates the arithmetic you came to do.

So: **generate code.** One straight-line function, state in locals, optimiser working across node boundaries.

We built it

Ran the wiring from `build.rs`, walked the graph, emitted a runner.

> **It failed three ways — and the third told us why.**

WHY A REWRITE, NOT A FEATURE

The generator failed three ways

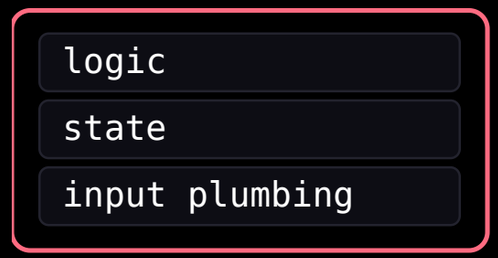
1. **It had to re-implement every node.** Semantics were welded to `RefCell` fields, so generated code could not *call* a node's cycle. Two `map`s, two `filter`s — nothing keeping them in step.
2. **Types and closures were already gone.** Wiring erased them. You cannot get `|x| x * 10` back out of a `Box<dyn Fn>`.
3. **Scheduling was inferred from names.** Nothing declared "I schedule myself", so the engine drove nodes off allowlists.

> **Node semantics were not *separately callable*.**

THE PATTERN

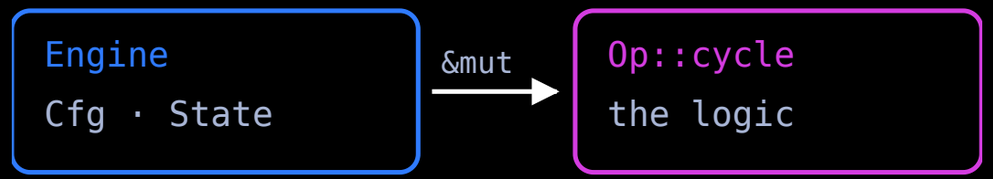
Inside-out

ONE OBJECT OWNS EVERYTHING



× nothing outside can call the logic

INSIDE-OUT



The tension

Encapsulation says bundle state with behaviour. Single responsibility says one reason to change. Follow the first far enough and you break the second.

The resolution

Encapsulate the **logic and the types**. Let **ownership** live outside.

> **Ownership moved out. Encapsulation didn't.**
>
> The op owns ***what*** and ***how***. The engine owns ***where*** and ***when***.

THE ONE DECISION EVERYTHING FOLLOWS FROM

Semantics became an associated function

```
trait Op {
    type Cfg;               // construction-time config; closures live here
    type State;             // engine-owned mutable state
    type In<'a>;            // typed inputs, passed in per cycle
    type Out;               // the produced value
    const ACTIVATION: Activation;

    fn cycle(cfg: &mut Self::Cfg, state: &mut Self::State,
             input: Self::In<'_>, ctx: &mut Ctx<'_>) -> Result<Tick<Self::Out>>;
}
```

> **Read what is *absent*. Nothing here says how state is stored, where inputs come from, or who calls it.**

THE SAME NODE, BOTH WAYS

The node changed completely. The wiring barely moved.

legacy — the node owns everything

```rust
#[node(active = [upstream], output = value: f64)]
impl MutableNode for ScaleStream {
    fn cycle(&mut self, _state: &mut GraphState)
        -> Result<bool>
    {
        self.value =
            self.upstream.peek_value() * self.factor;
        Ok(true)
    }
}
```

owns: its upstream handles, its state, its output slot

```rust
let count = ticker(Duration::from_millis(10))
    .count();
count.run(RunMode::RealTime, RunFor::Cycles(100))?;
```

now — the engine owns everything

```rust
#[op(build = scale, fluent)]
impl Op for Scale {
    type Cfg    = f64;              // config
    type State  = ();               // engine-owned
    type In<'a> = (&'a f64,);       // passed in
    type Out    = f64;
    const ACTIVATION: Activation = Activation::NONE;

    fn cycle(cfg: &mut f64, _state: &mut (),
             input: (&f64,), _ctx: &mut Ctx<'_>)
        -> Result<Tick<f64>>
    {
        Ok(Tick::Value(input.0 * *cfg))
    }
}
```

owns: nothing — cfg, state and inputs all arrive as arguments

```rust
let g = GraphBuilder::new();
let count = g.ticker(Duration::from_millis(10))
    .count();
g.build().run(RunMode::RealTime, RunFor::Cycles(100))?;
```

WHAT IT BUYS

Two engines, one copy of the semantics

Engine	Who owns state	Dispatch
Interpreted	the builder's slots	one dyn call per node
Compiled	locals in a generated fn	monomorphized, inlinable

One implementation — which is why they cannot drift.

…and you choose the boundary, not the engine

A **nested island** mounts the compiled emission as *one node* of an interpreted graph. Not a third engine — which is why it lands *between* the two.

THE FRONT DOOR

You write the wiring once, in plain Rust

```
wingfoil::nitro! {
    fn odds_evens(g: &GraphBuilder) -> Stream<String> {
        let count    = g.ticker(PERIOD).count();
        let is_even  = count.map(|i| i.is_multiple_of(2));
        let is_odd   = is_even.map(|b| !b);
        let odd_str  = count.filter(&is_odd).map(|i| format!("{i} is odd"));
        let even_str = count.filter(&is_even).map(|i| format!("{i} is even"));
        let labelled = odd_str.merge(&even_str);
        labelled
    }
}
```

Out comes `odds_evens::interpreted()`, `odds_evens::compiled()` and `odds_evens::nested()` — from the same tokens.

The wiring function stays **valid, reviewable Rust** either way. That property was deliberately never traded away: `count` is read three times, so it is a shared apex node — the macro derives a real DAG, not a chain.

…AND THIS IS WHAT IT BECOMES

One function. One local per node. No dyn.

```
let mut __k = Kernel::new(run_mode, run_for);

let mut __cfg_wf_anon_1   = PERIOD;
let mut __state_wf_anon_1 = __wf_op_ticker_seed_state(&__cfg_wf_anon_1);
let mut __v_wf_anon_1     = __wf_op_ticker_seed_value(&__cfg_wf_anon_1);
// … one cfg / state / value local per node, 11 nodes …

let mut __dirty = [false; 11usize];
while __k.begin_cycle(&mut __dirty) {
    let __t_wf_anon_1 = /* activation */ && {
        let mut __ctx = Ctx::new(&mut __k, 0usize);
        match __wf_op_ticker_cycle(&mut __cfg_wf_anon_1,
                                   &mut __state_wf_anon_1, (), &mut __ctx)
              .context("node 0 (wf_anon_1: ticker) cycle")? {
            Tick::Value(__val)  => { __v_wf_anon_1 = __val; true  }
            Tick::Silent(__val) => { __v_wf_anon_1 = __val; false }
            Tick::Quiet         => false,
        }
    };
    // … 10 more, emitted in topological order …
}
```

abridged · elisions marked · all 1,730 lines committed verbatim beside the example:
github.com/wingfoil-io/wingfoil → examples/core/dual_mode/expanded/

BOTH ENGINES ARE IN THE BINARY

Flip tiers without a rebuild

```
$ WINGFOIL_TIER=interpreted dual_mode
0.000_000 odds/evens "1 is odd"
0.010_000 odds/evens "2 is even"
0.020_000 odds/evens "3 is odd"
   …
ran 10 cycles on the interpreted tier
```

```
$ WINGFOIL_TIER=compiled dual_mode
0.000_000 odds/evens "1 is odd"
0.010_000 odds/evens "2 is even"
0.020_000 odds/evens "3 is odd"
   …
ran 10 cycles on the compiled tier
```

> **Same binary. Same output. Different engine.**

Develop interpreted — it honours the tracing span features and supports dynamic graph surgery, neither of which the monomorphized tiers have. **Deploy compiled.** `Tier::default()` falls back to the build profile, so the usual workflow needs no call-site change at all.

RESULTS

The numbers

~0.3 ns

per node cycle, compiled
~12 ns interpreted

4.4–37×

compiled vs interpreted

0.56–0.84×

new interpreted vs the *old* engine

Workload	Nodes	legacy	interpreted	compiled	nested
dense_chain	37	7.95 ms	6.64 ms	**187 µs**	931 µs
fanout	103	17.05 ms	11.99 ms	**324 µs**	1.43 ms
fan_in_256	260	38.08 ms	31.60 ms	**2.55 ms**	3.10 ms
sparse_wide	781	3.07 ms	2.01 ms	**355 µs**	896 µs

Shared 4-core VM — read the ratios, not the times. Eight workloads; four shown.

EXTENSIBILITY

Your op is not a second-class citizen

```
#[op(build = my_thing)]          // on your Op impl, in your crate
impl Op for MyThing { … }        // → the interpreted builder method
                                 // → the nitro! forwarders compiled emission uses
```

> **Within 2.4% of a built-in on the compiled path — no edit to the macro crate, no table to register in.**

PART THREE

What's next

Down the stack, up the stack,
and where we could use help.

DOWN THE STACK

Closer to the metal

1. **Core pin** — the graph thread on an isolated core.
2. **Kernel bypass** — Onload, then ef_vi/DPDK. This is the one that buys the wire-to-trade number we currently decline to claim.
3. **Zero-allocation I/O** — no allocator on the hot path.
4. **Verilog / FPGA** — the graph *as* gateware, with the backtest as its testbench.

> **Same graph definition. Further down.**

UP THE STACK

A platform, not just an engine

The missing layer

Simulated exchange · OMS / EMS · position keeping · risk · trade booking · venue adapters

None of it is a new kind of thing

Position keeping is a *fold*. Risk is a *filter*. PnL is a *join*. The simulated venue is just an *op*.

HELP WANTED

We'd love contributors

- Around **20 people** so far.
- **Good first issues** that really are — each names the file to change and code to copy.
- The bigger projects are separable, and each links to a design rather than a blank page.

Also genuinely useful

Where the docs lost you. The benchmark you don't believe. A project wingfoil would suit — or plainly wouldn't.

THANK YOU

wingfoil

> **If you work in financial services — or want to build a system to trade your own account — please do reach out.**

github.com/wingfoil-io/wingfoil · Apache-2.0

`cargo add wingfoil` · `pip install wingfoil`

hello@wingfoil.io · Discord · GitHub Discussions